SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA0606 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

Annexure A

Experiments

Code of Conduct:

I **Savitha R / 192421368** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SAVITHA R

1. Naïve Bayes Classifier Experiment

Concept

Naïve Bayes is a probabilistic classification algorithm based on **Bayes' theorem**. It assumes that the features are conditionally independent given the class.

Bayes' Theorem:

$P(C|X) = [P(X|C) \times P(C)] / P(X)$

For multiple features:

$P(C|X) \propto P(C) \times \prod_i P(X_i|C)$

Experiment

Use a real-world dataset such as the **Iris dataset**. Divide the data into training and testing sets. Train a Naïve Bayes classifier using different feature sets.

For example:

- Feature set 1: Sepal Length, Sepal Width
- Feature set 2: Petal Length, Petal Width
- Feature set 3: All four features

Evaluate each model using:

Accuracy:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision:

Precision = TP / (TP + FP)

Recall:

Recall = TP / (TP + FN)

Laplace Smoothing

Without smoothing, a feature value that does not occur in the training data can result in probability **0**, making the complete posterior probability zero.

Laplace smoothing avoids this:

P(x|C) = [count(x,C) + 1] / [count(C) + k]

where (k) is the number of possible feature values.

Analysis

The independence assumption makes Naïve Bayes simple and fast. However, if features are strongly correlated, the model may double-count their influence and reduce accuracy. Laplace smoothing generally improves reliability when unseen feature combinations occur.

Conclusion

Naïve Bayes performs well with suitable feature selection and small datasets. **Feature independence, preprocessing, and Laplace smoothing directly affect probability estimation and classification performance.**

2. MLE vs Bayesian Estimation

Concept

Maximum Likelihood Estimation (MLE) estimates parameters by selecting the values that maximize the likelihood of observing the given data.

$$\hat{\theta}\_ \text{MLE} = \arg\max \_\theta\ P(D|\theta)$$

For a normal distribution, MLE estimates:

$$\mu = (1/n)\ \Sigma_{i=1}^{n}\ x_i$$

and

$$\sigma^2 = (1/n)\ \Sigma_{i=1}^{n}\ (x_i - \mu)^2$$

Bayesian estimation combines the **prior distribution** with observed data:

$$P(\theta|D) \propto P(D|\theta)\ P(\theta)$$

Experiment

Take a numerical dataset and divide it into:

- Small sample: 10 observations
- Medium sample: 50 observations
- Large sample: 100 or more observations

Calculate mean and variance using MLE. Then assume a prior distribution for the parameters and calculate the Bayesian estimate.

Analysis

Sample Size	**MLE**	**Bayesian Estimation**
Small	More influenced by sample data	Prior has stronger influence
Medium	More stable	Data and prior both influence result
Large	Highly reliable	Data dominates the prior

With a small dataset, Bayesian estimation can provide more stable estimates because prior knowledge is incorporated. As the sample size increases, the effect of the prior becomes smaller and both methods tend to produce similar results.

Conclusion

MLE depends mainly on observed data, while Bayesian estimation combines **data and prior knowledge**. Bayesian estimation is particularly useful when the sample size is small or prior information is reliable.

3. Bayesian Belief Network Experiment

Concept

A **Bayesian Belief Network (BBN)** is a directed acyclic graph representing probabilistic relationships between variables.

For example:

Weather → Traffic → Late for College

The joint probability can be represented as:

$$P(W,T,L) = P(W)P(T|W)\ P(L|T)$$

Experiment

Construct a network with variables such as:

- Weather
- Traffic
- Accident
- Late

Assign conditional probability tables (CPTs).

For example:

$$P\ (Traffic=High\ |Rain) = 0.8$$

Calculate the probability of being late under different evidence conditions.

Evidence Experiments

Case 1: No evidence

Calculate the prior probability of being late.

Case 2: Evidence = Rain

Update the probability:

$$P\ (\text{Late}|\ \text{Rain})$$

Case 3: Evidence = Rain and Accident

Calculate:

$$P\ (\text{Late}\ |\text{Rain, Accident})$$

Then modify the conditional probabilities and observe how the posterior probability changes.

Analysis

Strong dependencies cause changes in one variable to influence connected variables. If the probability of high traffic given rain is increased, the posterior probability of being late also increases.

The BBN handles uncertainty by combining prior probabilities, conditional probabilities, and observed evidence.

Conclusion

Bayesian Belief Networks provide an effective way to represent **dependencies and uncertainty**. Changes in evidence or conditional probabilities directly affect posterior probabilities.

4. Candidate Elimination and Hypothesis Space

Concept

Candidate Elimination is a concept learning algorithm that maintains two boundaries:

- **S boundary:** Most specific hypotheses
- **G boundary:** Most general hypotheses

Together they represent the current **version space**.

The algorithm updates these boundaries whenever a positive or negative training example is received.

Experiment

Use a dataset such as **Play Tennis** with attributes:

- Outlook
- Temperature
- Humidity
- Wind

and target:

- Play = Yes/No

Perform experiments by varying:

1. Number of attributes
2. Number of training examples
3. Size of hypothesis space

Analysis

When the number of attributes increases, the hypothesis space generally becomes larger. Therefore, more training examples may be required to identify the correct hypothesis.

With fewer examples, many hypotheses may remain possible, resulting in poor generalization.

With more examples, the version space becomes smaller and the algorithm can converge toward consistent hypotheses.

Finite vs Infinite Hypothesis Space

Finite hypothesis space:

- Limited number of possible hypotheses
- Easier to analyze
- Better control over learning complexity

Infinite hypothesis space:

- Potentially unlimited hypotheses
- Requires stronger assumptions or regularization
- Greater risk of poor generalization

The mistake-bound model measures the maximum number of mistakes a learning algorithm may make before learning the correct concept.

Conclusion

Increasing the hypothesis space generally increases the complexity of learning. More representative training examples can improve **convergence, generalization, and learning efficiency**.

5. Decision Tree Classifier Experiment

Concept

A Decision Tree is a supervised learning algorithm that recursively divides data based on feature values. Two common splitting criteria are
Entropy/Information Gain and **Gini Index**.

Entropy:

$$\text{Entropy}(S) = -\Sigma_i \, p_i \log_2(p_i)$$

Information Gain:

$$IG = \text{Entropy}(\text{parent}) - \Sigma_v \, (|S_v| / |S|) \, \text{Entropy}(S_v)$$

Gini Index:

$$\text{Gini} = 1 - \Sigma_i \, p_i^2$$

Experiment

Use a real-world dataset such as the **Iris dataset**.

1. Split the dataset into training and testing sets.
2. Train one tree using **Entropy**.
3. Train another using **Gini Index**.
4. Compare their accuracy and F1-score.
5. Repeat using different tree depths.
6. Apply pruning and compare the results.

F1-score is:

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Sample Analysis

Configuration	Training Performance	Test Performance
Deep Tree	Very high	May decrease
Shallow Tree	Moderate	Better generalization
Entropy	Good	Depends on dataset
Gini	Good	Depends on dataset
Pruned Tree	Slightly lower training	Usually better test performance

Effect of Tree Depth

A very deep tree can memorize training data, causing **overfitting**. A very shallow tree may produce **underfitting**.

Pruning removes unnecessary branches and improves generalization.

Conclusion

Both Entropy and Gini can produce effective decision trees. Proper **tree depth, pruning, and cross-validation** are important for obtaining high accuracy and F1-score on unseen data.